

DM-J4310-2EC V1.1 – Cable-Drive Slack Removal

Low-Tension Winding: Theory and Implementation

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1 Problem Statement

Cable-driven robotic joints transmit force through tendons or Bowden cables. Before a cable-drive robot can be operated, every cable must be pre-tensioned: any slack must be wound out so that the cable lies straight and taut between the drum and the distal attachment point.

This document describes the problem of **automatic slack removal** using the DM-J4310-2EC motor as the cable drum actuator. The goal is:

1. Start with a wrapped but slack cable (the wrapping may be imprecise – the exact amount of slack is *not known* in advance).
2. Rotate the drum until the cable just becomes taut.
3. Stop immediately – do not overtension the cable or the distal joint.

Key constraints.

- The target shaft angle is *unknown* – position control requires a setpoint we do not have.
- Cable tension must not exceed the safe pre-tension limit (e.g. $\tau_{\text{pre}} \leq 0.5 \text{ N}\cdot\text{m}$ at the drum) to avoid cable damage or joint overload.
- The algorithm must be self-terminating: it must stop even if the cable is already taut at startup (zero-slack case).
- A timeout must exist to prevent indefinite rotation if a cable is missing or disconnected.

2 Control Mode Comparison

The DM-J4310-2EC exposes three families of control modes over CAN. Table 1 summarises their suitability for slack removal.

Table 1: CAN control modes vs. slack-removal suitability.

Mode	CAN ID	Needs setpoint?	Suitability
Position-Speed	0x100 + ID	Yes (p_{des} in rad)	Poor. Target position unknown; overshooting overtensions cable.
Speed	0x200 + ID	Yes (v_{des} in rad/s)	Moderate. Constant speed until stall; motor fights cable when taut; needs current monitoring to stop.
MIT Velocity	ID, $K_p = 0$, $K_d > 0$	$\dot{q}_{\text{des}}$ (rad/s)	Moderate. Similar to Speed mode; slightly easier to stop (velocity error grows at stall).
MIT Torque	ID, $K_p = 0$, $K_d = 0$	τ_{ff} (N·m)	Best. Motor self-stalls when cable load equals τ_{ff} ; no overshoot; magnitude directly sets pre-tension.

3 Recommended Strategy: MIT Torque Mode with Stall Detection

3.1 Physical intuition

In MIT torque mode ($K_p = K_d = 0$, $\tau_{\text{ff}} > 0$) the motor applies a constant torque to the drum. The torque balance on the drum shaft is:

$$J \ddot{q} = \tau_{\text{ff}} - \tau_{\text{friction}} - \tau_{\text{cable}}$$

where $\tau_{\text{cable}} = r \cdot T$ is the moment of cable tension T on a drum of effective radius r .

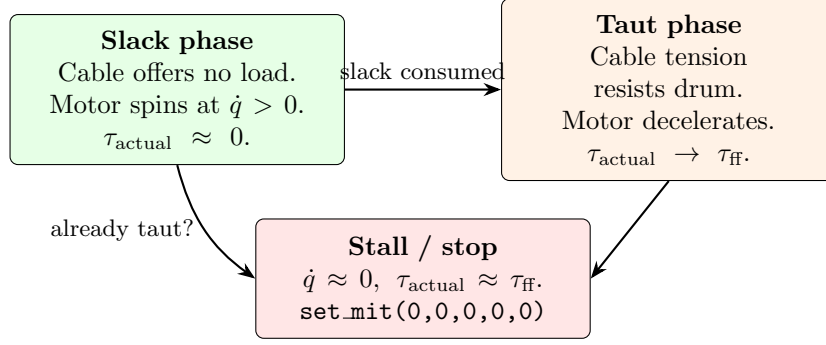


Figure 1: Phase diagram of the slack-removal procedure. MIT torque mode naturally transitions from the slack phase to a stall when cable tension equals τ_{ff} .

- **Slack:** $\tau_{\text{cable}} \approx 0$, so $\ddot{q} > 0$ and the motor accelerates until τ_{friction} limits it.
- **Transition:** as the last slack is wound onto the drum, T rises rapidly.
- **Taut:** τ_{cable} rises until $\tau_{\text{cable}} \geq \tau_{\text{ff}} - \tau_{\text{friction}}$, at which point $\ddot{q} \leq 0$ and the motor decelerates.
- **Stall:** $\dot{q} = 0$. The motor holds the cable at tension $T = \tau_{\text{ff}}/r$ and draws no further current (static stall).

This means τ_{ff} *directly controls the pre-tension* applied to the cable at the moment of stopping – which is the most important design parameter for cable-driven robots.

3.2 Stall detection criteria

Two independent signals are available from the DM-J4310-2EC CAN feedback: $\dot{q}$ (velocity, rad/s) and τ_{est} (estimated torque, N·m). Using both together avoids false triggers:

1. **Velocity criterion:** $|\dot{q}| < \varepsilon_v$ for a sustained duration Δt_{stall} . A single momentary slow-down (slack loop, knot passing) does not trigger a stop; the motor must be continuously stalled for Δt_{stall} .
2. **Torque criterion** (optional cross-check): $\tau_{\text{est}} > \tau_{\text{taut}}$, confirming actual load on the cable rather than a loose mechanical jam that stalls without tensioning.

The algorithm uses criterion 1 as the primary stop condition and criterion 2 as a guard to reject false stalls at startup (friction stall before cable is wound).

3.3 State machine

4 Parameter Selection

4.1 Winding torque τ_{ff}

The winding torque serves two simultaneous roles: it must be large enough to overcome bearing friction and cable bending stiffness, but small enough to set a safe cable pre-tension.

$$\tau_{\text{ff}} = r \cdot T_{\text{pre}} + \tau_{\text{friction}}$$

For the DM-J4310-2EC (TMAX = 10 N·m at output shaft):

Rule of thumb: start at $\tau_{\text{ff}} = 0.2 \text{ N·m}$ and increase until the motor reliably moves against the cable bending stiffness. Never exceed the cable's rated safe working load.

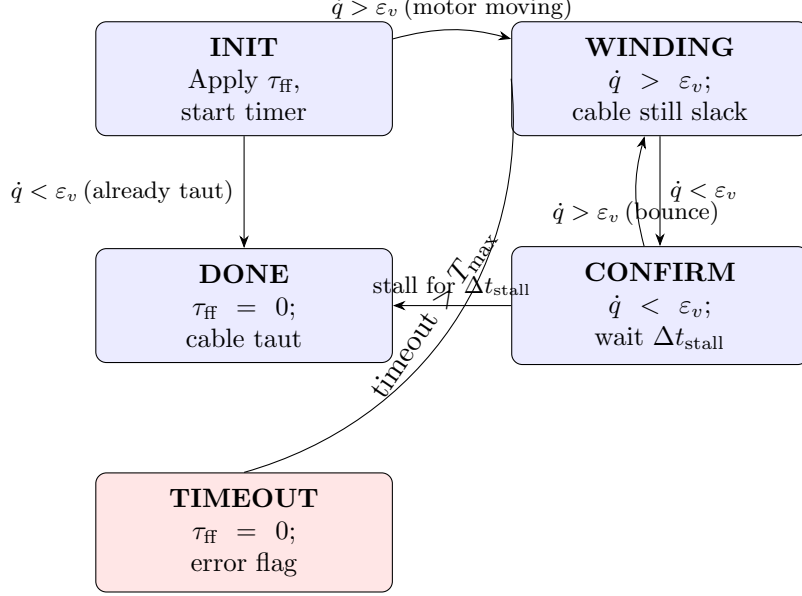


Figure 2: State machine for the slack-removal algorithm.

Table 2: Recommended τ_{ff} ranges.

Application	τ_{ff} (N·m)	Notes
Lightweight finger tendon ($r \approx 5$ mm, $T < 10$ N)	0.05–0.15	Very gentle; friction may dominate
Forearm/wrist cable ($r \approx 10$ mm, $T < 30$ N)	0.15–0.40	Typical cable-drive pre-tension
Upper-arm / large joint ($r \approx 20$ mm, $T < 50$ N)	0.40–1.00	Verify cable safe working load

4.2 Stall velocity threshold ε_v

Choose ε_v such that it is above measurement noise but well below any meaningful winding speed. For the J4310-2EC’s 14-bit encoder, the velocity resolution at 1 kHz is about 0.01 rad/s.

$$\varepsilon_v \approx 0.05\text{--}0.15 \text{ rad/s}$$

4.3 Stall confirmation time Δt_{stall}

A short burst of reduced velocity (e.g. a cable knot passing the drum) should not trigger a stop. Set:

$$\Delta t_{stall} \approx 0.25\text{--}0.5 \text{ s}$$

4.4 Safety timeout T_{max}

Protects against a disconnected or missing cable. Set to the maximum expected winding time:

$$T_{max} = \frac{L_{slack,max}}{r \cdot \dot{q}}$$

For example: 0.3 m slack, 10 mm drum radius, 1 rad/s winding speed $\Rightarrow T_{max} \approx 30$ s. The default in the test script is `TIMEOUT.S = 20.0`.

5 Implementation Notes for DM-J4310-2EC

5.1 Torque feedback caveat

The firmware’s torque estimate (τ_{est}) is derived from phase current and is noisy at low loads. Below $\approx 0.1 \text{ N}\cdot\text{m}$ the signal is not reliable for thresholding. Therefore the primary stop criterion uses *velocity* (reliable), and torque is used only as an optional cross-check when $\tau_{\text{ff}} \geq 0.3 \text{ N}\cdot\text{m}$.

5.2 Winding direction

The sign of τ_{ff} must match the cable-winding direction. Use a positive value if the cable winds when the output shaft rotates counter-clockwise when viewed from the encoder end (the default J4310 convention). Verify by commanding a tiny torque and observing which direction the motor moves.

5.3 Post-stall hold

After stall detection, the script has two options:

1. **Release** ($\text{TAU_FF} = 0$): motor becomes free. The cable may relax slightly due to elasticity. Use when the distal attachment is passive (spring-tensioned).
2. **Hold with low-stiffness position control** ($K_p \approx 5$, $K_d \approx 0.5$, $q_{\text{des}} = q_{\text{stall}}$): maintains the drum angle to preserve cable tension. Use when the cable must stay taut for subsequent operation.

The test script (Section 7) supports both via the `POST_STALL_HOLD` flag.

5.4 Output-shaft units

All CAN values are in **output-shaft units**: radians, rad/s, N·m. The 10:1 gearbox is transparent – the firmware uses the output-shaft encoder directly. Do *not* multiply torques or velocities by the gear ratio.

5.5 Winding speed (free-spin)

In MIT torque mode the winding speed is not commanded directly – it is determined by the balance of τ_{ff} minus friction. Typical free-spinning velocity for $\tau_{\text{ff}} = 0.3 \text{ N}\cdot\text{m}$ is 3–8 rad/s at the output shaft. If this is too fast for the cable mechanism, use MIT velocity mode instead (Section 6).

6 Alternative: MIT Velocity Mode with Torque Monitoring

If a controlled winding speed is required (e.g. a Bowden sheath that could kink), use MIT velocity mode ($K_p = 0$, $K_d > 0$, $\dot{q}_{\text{des}}$ small) and monitor the torque estimate:

- Set $\dot{q}_{\text{des}} = 0.5\text{--}2.0 \text{ rad/s}$ (slow, controlled).
- Monitor τ_{est} : when $\tau_{\text{est}} > \tau_{\text{taut}}$ for Δt_{stall} , cable is taut.
- Stop and optionally switch to position hold.

Limitation: the velocity loop fights the cable when taut, potentially applying larger forces than needed. A low K_d (0.3–0.8) reduces this.

7 Test Script: 11_cable_wind_to_tension.py

The script `tests/damiao/11_cable_wind_to_tension.py` implements the state machine of Section 3.3 using the `DamiaoBus` context manager.

7.1 Configurable parameters (top of script)

```
# ---- tuning -----
TAU_FF      = 0.3      # N.m winding torque (positive = CCW drum)
V_STALL_EPS = 0.10     # rad/s below this -> treat as stalled
T_STALL_CONFIRM = 0.35 # s must be stalled this long to confirm
TAU_TAUT_MIN = 0.15    # N.m torque must exceed this during stall
TIMEOUT_S   = 20.0     # s abort if not taut within this time
WINDING_HZ  = 100      # Hz control loop rate
POST_STALL_HOLD = True  # True -> hold drum angle after taut
KP_HOLD     = 5.0      # spring for post-stall hold (if enabled)
KD_HOLD     = 0.5      # damping for post-stall hold
```

7.2 State machine output

```
[cable_wind] State: INIT | q=+0.00 rad dq=+0.00 rad/s tau=+0.00 Nm
[cable_wind] State: WINDING| q=+0.12 rad dq=+3.42 rad/s tau=+0.04 Nm
...
[cable_wind] State: CONFIRM| q=+2.18 rad dq=+0.02 rad/s tau=+0.28 Nm
[cable_wind] DONE: cable taut at q=+2.18 rad after 0.72 s.
[cable_wind] Holding drum angle q=+2.18 rad with Kp=5.0, Kd=0.5
```

7.3 Safety notes

- **Power on the motor before running the script.** The script enables the motor on entry.
- **Keep the cable force path clear.** Ensure the distal attachment is secure before running.
- **Ctrl+C** will stop the motor immediately (zero torque sent in the `finally` block).
- **Do not set TAU_FF above the cable's safe working load divided by the drum radius.**

8 Quick Reference

Table 3: Parameter quick reference for `11_cable_wind_to_tension.py`.

Parameter	Default	Unit	When to change
TAU_FF	0.3	N·m	Adjust for cable pre-tension target
V_STALL_EPS	0.10	rad/s	Increase in noisy environments
T_STALL_CONFIRM	0.35	s	Increase if false triggers occur
TAU_TAUT_MIN	0.15	N·m	Raise if τ_{ff} is large
TIMEOUT_S	20.0	s	Set to max expected winding time
POST_STALL_HOLD	True	bool	False = release after taut
KP_HOLD	5.0	–	Stiffen to resist distal forces
KD_HOLD	0.5	–	Increase to damp hold oscillations